# Euclid's Elements — Book XI
# Proposition 9

Formal Reconstruction — Technical Documentation

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CGJteamLab

*If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.*

## 1 Introduction

Proposition XI.9 is the spatial counterpart of Euclid I.30. It asserts that if two lines are each parallel to a third line, then they are parallel to one another, provided that the three lines are not all contained in one plane.

This extra non-coplanarity clause is the essential new feature. In a plane, Euclid I.30 already proves transitivity of parallelism. In space, however, the theorem must distinguish two cases:

- the planar case, where the three lines lie in one plane and I.30 is the relevant model;
- the genuinely spatial case, where the three lines are distributed across different planes and the conclusion must be rebuilt from 3D incidence.

The current formal proof treats XI.9 as a theorem of synthetic spatial incidence plus one plane-local use of the Euclidean parallel axiom. Contrary to the historical proof, it does not proceed through a chain of perpendicular constructions.

The production file is

```
CGJteamLab/Proposition11_9.lean
```

and XI.9 introduces two proposition-local helpers:

```
hilbert_XI9_point_on_second_line_off_first
hilbert_XI9_planes_eq_of_common_line_and_external_point
```

## 2 Euclid's Statement

Euclid states:

> If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.

In the present library, spatial parallelism is represented by

```
HilbertSpaceLinesParallel Geo l m
```

which packages:

- a plane containing both lines, and
- ambient disjointness of the two lines.

Thus the formal conclusion of XI.9 is not merely "coplanar with the same direction" but the full structured statement that $l$ and $m$ are parallel lines in the project's spatial sense.

The phrase "if they do not lie in the same plane with it" is formalized as

```
Not (exists omega : S.Plane,
  HilbertLineInPlane Geo l omega /\
  HilbertLineInPlane Geo m omega /\
  HilbertLineInPlane Geo n omega)
```

That is: there is no single plane containing all three lines $l, m, n$. This is the exact ambient reading of Euclid's proviso.

## 3 Classical Configuration

We are given three lines $l, m, n$ such that

$$l \parallel n, \qquad m \parallel n,$$

and the three are not all coplanar.

Let $\sigma$ be a plane containing $l$ and $n$, and let $\tau$ be a plane containing $m$ and $n$. Because all three lines are not contained in one plane, these two carrier planes are distinct:

$$\sigma \neq \tau.$$

The formal proof then chooses a point $B$ on $m$ not lying on $l$, forms the plane $\omega$ through $l$ and $B$, and studies the intersection

$$r = \omega \cap \tau.$$

The line $r$ passes through $B$. Inside $\tau$, both $r$ and $m$ pass through $B$ and are parallel to $n$. Euclidean uniqueness in the plane $\tau$ forces

$$r = m.$$

Hence both $l$ and $m$ lie in $\omega$. The final step proves they are disjoint, so

$$l \parallel m.$$

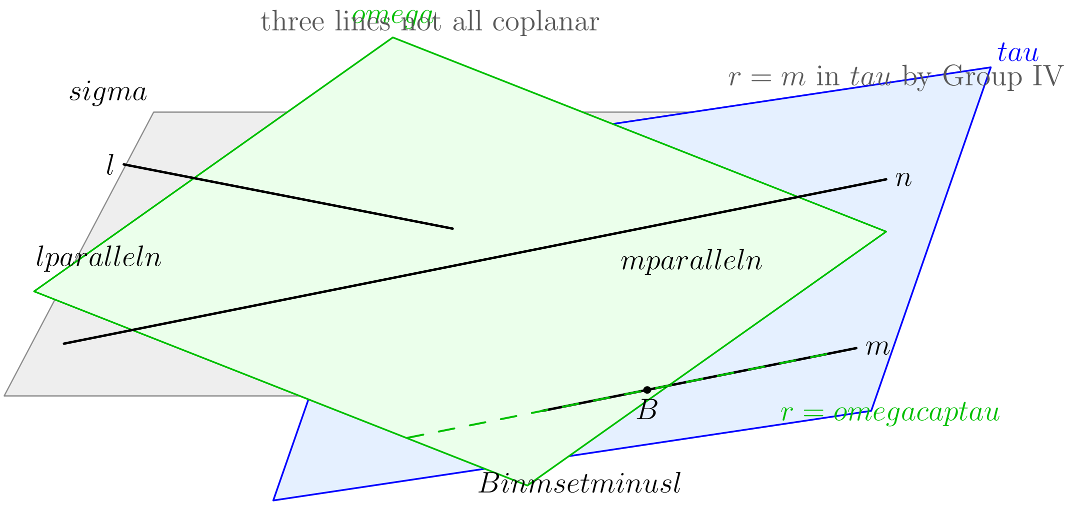


Figure 1: A schematic picture for Proposition XI.9. The line $n$ lies in both carrier planes $\sigma$ and $\tau$. The line $l$ lies in $\sigma$, the line $m$ lies in $\tau$, and the three lines are not all coplanar. Choosing $B \in m \setminus l$, the plane $\omega$ through $l$ and $B$ intersects $\tau$ in a line $r$. Since both $r$ and $m$ pass through $B$ and are parallel to $n$ inside $\tau$, Group IV gives $r = m$.

## 4 Euclid's Classical Proof Pattern

Euclid's printed proof is not organized in this affine way. Its characteristic strategy is to convert the parallel problem into a perpendicular problem. Schematically, the classical route is:

$$\text{I.11 in two planes} \longrightarrow \text{common perpendiculars} \longrightarrow \text{XI.4} \longrightarrow \text{XI.8} \longrightarrow \text{XI.6} \longrightarrow \text{XI.9}.$$

In more explicit local terms:

1. in suitable planes, erect lines perpendicular to the common line;
2. use XI.4 to pass from right-angle data in one plane to perpendicularity to the plane;
3. use XI.8 to transport plane-perpendicularity across parallels;
4. use XI.6 to conclude that the resulting perpendiculars are parallel;
5. conclude that the original lines are parallel.

This is entirely reasonable in Euclid's sequence, since Book XI is gradually building a theory of perpendicularity and parallelism for lines and planes.

Nevertheless, from the standpoint of the current formal library, this is not the most efficient reconstruction. The actual theorem can be proved more directly from spatial incidence and planar parallel uniqueness.

# 5 Why the Formal Reconstruction Takes a Different Route

The project already contains a precise synthetic notion of ambient line parallelism, a theorem on the intersection of two planes, and a plane-local Euclidean parallel uniqueness axiom.

Once these are available, XI.9 can be reconstructed with a cleaner structure:

- first create the target coplanarity of $l$ and $m$;
- then prove their disjointness;
- package both facts as `HilbertSpaceLinesParallel Geo l m`.

This proof is still fully synthetic. It uses no coordinates, vectors, or scalar products. The difference is conceptual, not analytic: the formal proof exposes XI.9 as a theorem about how distinct planes and parallel lines fit together in 3-space.

# 6 Proposition-Local Helper 1

The first local helper is

```
theorem hilbert_XI9_point_on_second_line_off_first
    ...
    (l m : Geo.Line)
    (hlm : Ne l m) :
    exists P : Geo.Point,
      H.OnLine P m /\
      Not (H.OnLine P l)
```

Its content is elementary but useful: if two lines are distinct, then the second line has a point not lying on the first.

The proof uses two points on $m$. If both were also on $l$, line uniqueness would force $m = l$, contradiction.

This helper is a pure incidence lemma and abstracts away a routine but repeatedly needed argument.

# 7 Proposition-Local Helper 2

The second helper is

```
theorem hilbert_XI9_planes_eq_of_common_line_and_external_point
    ...
    (l : Geo.Line)
    (P : Geo.Point)
    (hPl : Not (H.OnLine P l))
    (pi rho : S.Plane)
    (hlpi : HilbertLineInPlane Geo l pi)
    (hlrho : HilbertLineInPlane Geo l rho)
    (hPpi : S.OnPlane P pi)
    (hPrho : S.OnPlane P rho) :
    pi = rho
```

This is the uniqueness clause of the theorem "a line and an external point determine a unique plane," extracted into a convenient reusable form.

Its role in XI.9 is structural. Whenever two planes both contain the same line and the same point outside that line, the planes coincide. This is the key contradiction engine used twice in the proof:

- first to show that the auxiliary intersection line $r$ is disjoint from $n$;
- then to show that the final target lines $l$ and $m$ are disjoint.

## 8 Formal Statement

The public theorem is:

```
theorem euclid_proposition_11_9
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
    [HSE : HilbertSpaceEuclidean Geo]
    (l m n : Geo.Line)
    (hParallelLN : HilbertSpaceLinesParallel Geo l n)
    (hParallelMN : HilbertSpaceLinesParallel Geo m n)
    (hNoCommonPlane :
      Not (exists omega : S.Plane,
        HilbertLineInPlane Geo l omega /\
        HilbertLineInPlane Geo m omega /\
        HilbertLineInPlane Geo n omega)) :
    HilbertSpaceLinesParallel Geo l m
```

The assumptions separate clearly into three layers:

- spatial incidence and plane structure;
- spatial order/congruence, needed through the Group IV interface;
- the Euclidean parallel uniqueness axiom `HilbertSpaceEuclidean Geo`.

# 9 Proof Architecture of the Reconstruction

The proof has six main steps.

### Step 1: unpack the two given parallels

From

$$l \parallel n, \qquad m \parallel n,$$

we obtain planes $\sigma$ and $\tau$ such that

$$l, n \subset \sigma, \qquad m, n \subset \tau,$$

with the corresponding disjointness facts

$$l \cap n = \varnothing, \qquad m \cap n = \varnothing.$$

### Step 2: prove $\sigma \neq \tau$ and $l \neq m$

If $\sigma = \tau$, then the three lines would lie in one plane, contrary to the explicit hypothesis `hNoCommonPlane`. Likewise, if $l = m$, then again all three would lie in one plane.

### Step 3: choose $B \in m \setminus l$ and form $\omega$

Using `hilbert_XI9_point_on_second_line_off_first`, choose a point $B$ on $m$ which does not lie on $l$. Then the theorem

```
hilbert_plane_through_line_and_external_point
```

produces a plane $\omega$ containing $l$ and $B$.

### Step 4: intersect $\omega$ and $\tau$

The planes $\omega$ and $\tau$ are distinct, and both contain $B$. By XI.3 in its strengthened Hilbert form, they intersect in a line $r$ through $B$:

$$r = \omega \cap \tau.$$

### Step 5: identify $r$ with $m$

The crucial point is that $r$ is disjoint from $n$. For if a point $P$ belonged to both $r$ and $n$, then $P$ would lie in both $\omega$ and $\sigma$, while being outside $l$. Since both planes also contain $l$, helper 2 would force

$$\sigma = \omega,$$

contradiction.

Thus in the plane $\tau$, both lines $r$ and $m$:

- pass through the point $B$, and
- are disjoint from the line $n$.

By plane-local Euclidean uniqueness of a parallel through a point,

$$r = m.$$

Hence $m \subset \omega$, and the target coplanarity of $l$ and $m$ is established.

### Step 6: prove $l$ and $m$ are disjoint

Suppose a point $P$ lay on both $l$ and $m$. Since $l$ is disjoint from $n$, this point is outside $n$. But $P$ lies in $\sigma$ (because $P \in l \subset \sigma$) and in $\tau$ (because $P \in m \subset \tau$). Since both planes also contain $n$, helper 2 forces

$$\sigma = \tau,$$

again contradicting the non-coplanarity assumption. Therefore

$$l \cap m = \varnothing.$$

Since $l$ and $m$ lie in the common plane $\omega$ and are disjoint, the packaged conclusion follows:

$$l \parallel m.$$

## 10 Formal DAG

The actual dependency chain of the current reconstruction is:

```
hParallelLN, hParallelMN, hNoCommonPlane
        |
        +-- hilbert_XI9_point_on_second_line_off_first
        |
        +-- hilbert_plane_through_line_and_external_point
        |
        +-- hilbert_plane_intersection_line   (XI.3 infrastructure)
        |
        +-- HilbertSpaceEuclidean.parallel_unique_in_plane
        |
```

```
        +-- hilbert_XI9_planes_eq_of_common_line_and_external_point
        |
        v
euclid_proposition_11_9
```

This should be contrasted with the historical local chain:

```
I.11
  -> XI.4
  -> XI.8
  -> XI.6
  -> XI.9
```

The formal DAG is not less synthetic. It simply exposes the spatial incidence content more directly.

## 11 Spatial / Planar Architecture Used

The theorem depends on the following ambient structure:

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
```

It also reuses the previously developed spatial theorems:

```
hilbert_plane_through_line_and_external_point
hilbert_plane_intersection_line
```

No coordinates, vectors, affine spaces, or scalar products are introduced. No induced `PlaneGeo` slice is required in the public proof, even though the decisive Euclidean axiom is conceptually plane-local.

## 12 Role of Group IV

XI.9 uses the Euclidean parallel axiom exactly once, in the form

```
HilbertSpaceEuclidean.parallel_unique_in_plane
```

applied in the plane $\tau$.

This is the precise place where Euclidean strength enters. Everything before that point is incidence-only: choose a point off a line, construct a plane, intersect two planes, and compare two candidate planes by uniqueness.

The proof therefore gives a clean conceptual reading of XI.9:

$$\text{3D incidence geometry} + \text{one planar uniqueness step} \implies \text{spatial transitivity of parallelism under Euclid's prov}$$

## 13 Hidden Formal Data

Several facts that are implicit in the classical picture must be made explicit in Lean.

### The carrier planes are proof objects

The hypotheses `hParallelLN` and `hParallelMN` do not merely say "parallel" in a bare relational sense. They supply witness planes $\sigma$ and $\tau$.

### The non-coplanarity clause is genuinely global

The condition `hNoCommonPlane` is not a statement about one chosen pair of lines; it quantifies over all candidate planes $\omega$.

### The point $B$ is not diagrammatic decoration

The point off $l$ but on $m$ is the key to constructing the new plane $\omega$ that will eventually contain both target lines.

### Disjointness must be proved twice

The proof must explicitly certify both

$$r \cap n = \varnothing \qquad \text{and} \qquad l \cap m = \varnothing.$$

In a traditional diagram these facts often look visually obvious, but in a synthetic proof they are independent obligations.

## 14 Axiomatic Status

The audit file contains

```
import CGJteamLab.Proposition11_9

#print axioms Geometry.hilbert_XI9_point_on_second_line_off_first
#print axioms Geometry.hilbert_XI9_planes_eq_of_common_line_and_external_point
#print axioms Geometry.euclid_proposition_11_9
```

Lean reports:

```
'Geometry.hilbert_XI9_point_on_second_line_off_first'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI9_planes_eq_of_common_line_and_external_point'
does not depend on any axioms

'Geometry.euclid_proposition_11_9'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

This is an excellent outcome for XI.9.

- No new project-specific global geometric axiom constant is introduced.
- The public theorem does not depend on `Geometry.bookZero_nullSegment1`.
- The explicit typeclass hypothesis `[HSE : HilbertSpaceEuclidean Geo]` remains a genuine geometric assumption, but it is not counted by `#print axioms` because it is an explicit theorem parameter rather than a hidden global constant.

At this checkpoint:

| | |
|---|---|
| New proposition-specific geometric axiom | no |
| New proposition-local helpers | yes (two) |
| New reusable general interface theorem | no |
| Hilbert Group IV / Euclidean parallelism | yes |
| Continuity | no |
| PlaneGeo required in public proof | no |
| Coordinates / vectors / scalar product | no |
| `sorry`/`admit` | no |
| Public theorem compiles | yes |
| Axiom audit completed | yes |

## 15 Mathematical Significance of the Reconstruction

XI.9 is a good example of the difference between the order of exposition in the *Elements* and the most transparent dependency structure of the formal theory.

Euclid's proof moves through perpendicularity because Book XI is organized as a gradual synthetic development of the geometry of lines and planes. The modern formal system, however, already has explicit infrastructure for planes through lines, intersections of planes, and plane-local uniqueness of parallels.

Once those pieces are isolated, the theorem turns out to be structurally about *coplanarity creation* and *disjointness control*. The core problem is not to manufacture right angles, but to produce the unique plane in which the two target lines must lie.

In this sense XI.9 is one of the first propositions of Book XI where the formal reconstruction becomes conceptually more informative than the classical proof: it reveals the theorem as a hybrid of spatial incidence and planar Euclidean uniqueness.

## 16 Conclusion

Proposition XI.9 states that if two lines are each parallel to a third line, and the three are not all contained in one plane, then the first two are parallel to one another.

The present Lean proof establishes this in a direct synthetic way:

$$l \parallel n, \quad m \parallel n, \quad \text{no common plane for } l, m, n \Longrightarrow l \parallel m.$$

The proof proceeds by choosing a point on $m$ outside $l$, constructing a plane through $l$ and that point, intersecting it with the plane containing $m$ and $n$, and invoking Euclidean uniqueness of a parallel through a point. A second plane-uniqueness argument then supplies the required disjointness of $l$ and $m$.

Thus XI.9 is formalized not as a proposition about perpendicular gadgets, but as a structural theorem about three lines distributed across several planes. It remains fully synthetic, uses Group IV at exactly one point, and achieves a very clean axiom profile.